

```

117.     # Generate signals
118.     data['Signal'] = 0
119.     data.loc[data['SMA_Short'] > data['SMA_Long'], 'Signal'] = 1
120.
121.     # Run backtest
122.     backtest_result = backtest_strategy(data)
123.
124.     # Evaluate results
125.     metrics = evaluate_backtest(backtest_result)
126.
127.     # Store results
128.     result = {
129.         'short_ma': short_ma,
130.         'long_ma': long_ma,
131.         **metrics
132.     }
133.     final_results.append(result)
134.
135.     # Convert results to DataFrame
136.     results_df = pd.DataFrame(final_results)
137.
138.     # Sort by Sharpe Ratio (or another preferred metric)
139.     results_df = results_df.sort_values('Sharpe Ratio', ascending=False)
140.
141.     return results_df
142.

```

Example usage:

```

1. # Define parameter bounds
2. param_bounds = {
3.     'short_ma': (5, 50), # Min and max values for short MA
4.     'long_ma': (50, 200) # Min and max values for long MA
5. }
6.
7. # Perform genetic algorithm optimization
8. results = genetic_algorithm_optimization('AAPL', '2018-01-01', '2023-01-01',
9.                                           param_bounds, population_size=30, generations=10)
10.
11. # Display top 5 parameter combinations

```